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2014.0 RANGE ROVER (LG), 303-14

ELECTRONIC ENGINE CONTROLS - V8 S/C 5.0L PETROL

PRINCIPLES OF OPERATION

For a detailed description of the electronic engine control system and operation, refer to the relevant Description and Operation section of the workshop manual. REFER to: Electronic Engine Controls (303-14D, Description and Operation).

INSPECTION AND VERIFICATION

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault and may also cause additional faults in the vehicle being checked and/or the donor vehicle.

NOTE:

Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.

1. Verify the customer concern.

1. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Engine oil level ▪ Cooling system coolant level ▪ Fuel level ▪ Fuel contamination/grade/quality ▪ Fuel leaks ▪ Front end accessory drive belt ▪ Air filter and induction hoses ▪ Exhaust system ▪ Charge air intake system	▪ Fuses ▪ Wiring harness ▪ Electrical connector(s) ▪ 5 volt sensor supply ▪ Sensor(s) ▪ Engine control module ▪ Transmission control module

▪ Supercharger	

1. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

1. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for Diagnostic Trouble Codes (DTCs) and refer to the DTC Index.

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
Engine cranks, but does not fire	▪ Engine breather system disconnected /restricted ▪ Ignition system ▪ Fuel system ▪ Electronic engine control	▪ Ensure the engine breather system is free from restriction and is correctly installed. Check for ignition system, fuel system and electronic engine control DTCs and refer to the relevant DTC Index
Engine cranks and fires, but will not start	▪ Evaporative emissions purge valve ▪ Fuel pump ▪ Spark plugs ▪ HT short to ground (tracking) check rubber boots for cracks /damage ▪ Ignition system	▪ Check for evaporative emissions, fuel system and ignition system related DTCs and refer to the relevant DTC Index
Difficult cold start	▪ Engine coolant level /anti-freeze content ▪ Battery ▪ Electronic engine controls	▪ Check the engine coolant level and condition. Ensure the battery is in a fully charged and serviceable condition. Check for electronic engine controls, engine emissions, fuel system and evaporative emissions system related DTCs and refer to the relevant DTC Index

	- Exhaust gas recirculation valve stuck open - Fuel pump - Purge valve	
Difficult hot start	- Injector leak - Electronic engine control - Purge valve - Fuel pump - Ignition system - Exhaust gas recirculation valve stuck open	- Check for injector leak, install new injector as required. Check for electronic engine controls, evaporative emissions, fuel system, ignition system and engine emission system related DTCs and refer to the relevant DTC Index
Difficult to start after hot soak (vehicle standing, engine off, after engine has reached operating temperature)	- Injector leak - Electronic engine control - Purge valve - Fuel pump - Ignition system - Exhaust gas recirculation valve stuck open	- Check for injector leak, install new injector as required. Check for electronic engine controls, evaporative emissions, fuel system, ignition system and engine emission system related DTCs and refer to the relevant DTC Index
Engine stalls soon after start	- Breather system disconnected /restricted - Engine control module relay - Electronic engine control - Ignition system - Air intake system restricted - Air leakage - Fuel lines	- Ensure the engine breather system is free from restriction and is correctly installed. Check for electronic engine control, ignition system and fuel system related DTCs and refer to the relevant DTC Index. Check for blockage in air filter element and air intake system. Check for air leakage in air intake system

Engine hesitates /poor acceleration	▪ Fuel pressure, fuel pump, fuel lines ▪ Injector leak ▪ Boost air leakage ▪ Supercharger failure ▪ Electronic engine control ▪ Throttle motor ▪ Restricted accelerator pedal travel (carpet, etc) ▪ Ignition system ▪ Exhaust gas recirculation valve stuck open ▪ Transmission malfunction	▪ Check for fuel system related DTCs and refer to the relevant DTC Index. Check for injector leak, install new injector as required. Check for air leakage in air intake system. Check for correct supercharger operation. Ensure accelerator pedal is free from restriction. Check for electronic engine controls, ignition, engine emission system and transmission related DTCs and refer to the relevant DTC Index
Engine backfires	▪ Fuel pump /lines ▪ Air leakage ▪ Electronic engine controls ▪ Ignition system ▪ Sticking variable camshaft timing hub	▪ Check for fuel system failures. Check for air leakage in intake air system. Check for electronic engine controls, ignition system and variable camshaft timing system related DTCs and refer to the relevant DTC Index
Engine surges	▪ Fuel pump /lines ▪ Electronic engine controls ▪ Throttle motor ▪ Ignition system	▪ Check for fuel system failures. Check for electronic engine controls, throttle system and ignition system related DTCs and refer to the relevant DTC Index
Engine detonates	▪ Fuel pump /lines	▪ Check for fuel system failures. Check for air leakage in intake air system. Check for electronic engine controls and

/knocks	- Air leakage - Electronic engine controls - Sticking variable camshaft timing hub	variable camshaft timing system related DTCs and refer to the relevant DTC Index
No throttle response	- Electronic engine controls - Throttle motor	- Check for electronic engine controls and throttle system related DTCs and refer to the relevant DTC Index
Poor throttle response	- Breather system disconnected /restricted - Electronic engine control - Transmission malfunction - Traction control event - Boost air leakage	- Ensure the engine breather system is free from restriction and is correctly installed. Check for electronic engine controls, transmission and traction control related DTCs and refer to the related DTC Index. Check for air leakage in intake air system

PINPOINT TEST A : MISFIRE PINPOINT TEST	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
A1: CHECK FOR MISFIRE COUNTS	
Check the Powertrain control module (PCM) for related engine system DTCs and refer to relevant DTC index. If DTCs relating to any subsystem are present then these must be addressed before continuing with this pinpoint test e.g. Ignition system (including crank shaft sensor), Fuelling system (includes fuel system too lean/rich)	
	1 Using the Jaguar Land Rover Approved Diagnostic Equipment, run the **Powertrain - Misfire - Cylinder Identification** application
	Can a **persistent** misfire be confirmed **Yes** Proceed to next step **No** If the misfire is not present, ensure conditions required to replicate the misfire are understood and repeat the test Consider the following: DTC snapshot information: Engine temperature, Engine speed, Vehicle speed, Time since the engine was started Driving conditions when the misfire occurred
A2: VISUAL CHECK	

	1 Inspect the wiring harness for the ignition coils and fuel injectors. Ensure that the electrical connector pins are not backed out or corroded and the electrical connectors latch correctly
	Have any wiring harness or electrical connector faults been identified? **Yes** Rectify the fault as required Repeat test step 1 (CHECK FOR MISFIRE COUNTS) **No** Proceed to next step
A3: SWAP IGNITION COIL CYLINDER TO CYLINDER	
	1 Swap the ignition coil from the affected cylinder(s) with a coil from a known good cylinder (do not remove the spark plug)
	2 Repeat test step 1 (CHECK FOR MISFIRE COUNTS)
	Has the misfire changed cylinder position? **Yes** Check and install a new ignition coil as required Repeat test step 1 (CHECK FOR MISFIRE COUNTS) **No** Proceed to next step
A4: SWAP SPARK PLUG CYLINDER TO CYLINDER	
	1 Swap the spark plug(s) from the affected cylinder(s) with a spark plug from a known good cylinder
	2 Repeat test step 1 (CHECK FOR MISFIRE COUNTS)
	Has the misfire changed cylinder position? **Yes** Check and install new spark plug(s) as required Repeat test step 1 (CHECK FOR MISFIRE COUNTS) **No** Proceed to next step
A5: ELECTRICAL CHECKS (OSCILLOSCOPE REQUIRED)	
	1 Using an oscilloscope check the actuation signal from the Powertrain Control Module (PCM) to the ignition coil
	2 Compare the affected cylinder actuation signal for plausibility against known good cylinder actuation signals
	Is the actuation signal for the affected cylinder plausible? **Yes** Proceed to next step **No** Refer to the electrical circuit diagrams and check the affected cylinder ignition coil circuit for short circuit to ground, short circuit to power, open circuit, high resistance Inspect the electrical connectors for signs of water ingress and pins for damage and/or corrosion
A6: CHECK FUEL TRIMS	
	1 Using the Jaguar Land Rover Approved Diagnostic Equipment, run the **Powertrain - Adaptive Fuel Trim Display** application
	2 Check the short term and long term fuel trims
	Does the application show the values within tolerance? **Yes** Proceed to next step **No**

	Using the Jaguar Land Rover Approved smoke machine, check the air intake system and vacuum hoses for leakage, blockage and mechanical integrity Check and install new air intake or vacuum system components only as required Repeat test step 1 (CHECK FOR MISFIRE COUNTS)
A7: CHECK MASS AIR FLOW SENSOR	
	1 Using the Jaguar Land Rover Approved Diagnostic Equipment, check datalogger signal 0x0504 - Air Flow Sensor Bank 1
	2 Using the Jaguar Land Rover Approved Diagnostic Equipment, check datalogger signal 0x0505 - Air Flow Sensor Bank 2
	3 Compare the sensor values at idle and part load
	Are the sensor values comparable between bank 1 and bank 2 **Yes** Proceed to step 9 (SWAP FUEL INJECTORS CYLINDER TO CYLINDER) **No** Proceed to next step
A8: SWAP MASS AIR FLOW SENSOR	
	1 Swap mass air flow sensors bank to bank
	2 Using the Jaguar Land Rover Approved Diagnostic Equipment, check datalogger signal 0x0504 - Air Flow Rate From Mass Air Flow Sensor Bank 1
	3 Using the Jaguar Land Rover Approved Diagnostic Equipment, check datalogger signal 0x0505 - Air Flow Rate From Mass Air Flow Sensor Bank 2
	4 Compare the sensor values at idle and part load
	Are the sensor values comparable between bank 1 and bank 2 **Yes** Proceed to next step **No** Refer to the electrical circuit diagrams and check the mass air flow sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance Inspect connectors for signs of water ingress and pins for damage and/or corrosion Check for restricted air flow Check and install a new mass air flow sensor component only as required Repeat test step 1 (CHECK FOR MISFIRE COUNTS)
A9: SWAP FUEL INJECTORS CYLINDER TO CYLINDER	
	1 Swap the fuel injector(s) from the affected cylinder(s) with a fuel injector from a known good cylinder
	2 Repeat test step 1 (CHECK FOR MISFIRE COUNTS)
	Has the misfire changed cylinder position? **Yes** Check and install new fuel injector(s) Only the fuel injector for the corresponding DTC cylinder referenced should be replaced Repeat test step 1 (CHECK FOR MISFIRE COUNTS) **No** Proceed to next step
A10: CARRY OUT ENGINE COMPRESSION TEST	
	1 Where available, using the Jaguar Land Rover Approved Diagnostic Equipment, run the **Inline Diagnostic Unit 2 relative compression test** application, available on the following vehicles: ▪ **AJ126 and AJ133:** L405, L494, X152, X260, L319 MY14>, X250 MY13>, X351 MY13>

	■ For vehicles not covered by the diagnostic tool compression test, refer to TOPIx section **303-00 Cylinder Compression Test**
	Were the compression test values within limits? **Yes** proceed to next step **No** Carry out a cylinder leakage test to identify the area of compression loss e.g. cylinder or valve issue
A11: OTHER POSSIBLE CAUSES	
The items on the following list are less likely to have set the DTC(s) but should be considered as other possibilities to be checked	
	1 Other possible causes: ■ Crankshaft position sensor incorrect installation ■ Reluctor ring mechanical damage at the flywheel ■ Crank case ventilation system blockage or leakage
	Were any of the above issues identified? **Yes** Rectify the fault as required Repeat test step 1 (CHECK FOR MISFIRE COUNTS) **No** Reconfirm the customer issue, ensure conditions required to replicate the misfire are understood and repeat the test

DTC INDEX

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00.